ROYAL JORDANIAN AIR FORCE

Squadron Operations & Pilot Logbook System

COMMAND BRIEFING

A Single, Secure System for Every Squadron.

A purpose-built operations platform that replaces paper logbooks, scattered spreadsheets, and informal phone-tree updates with one command-controlled, encrypted, real-time picture of pilot readiness.

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00 Executive Summary

In one sentence: this system gives every commander a live, accurate picture of pilot hours, currency, risk and readiness across the Force — while making sure each commander only sees the squadrons they are authorised to see.

Today, pilot hours, currency dates, sortie cards, and risk-assessment forms live on paper, in personal notebooks, and in a different Excel file on every desk. Information is correct on the day it's written and wrong by the next morning. A wing commander asking "how many hours did Captain X fly last quarter?" waits hours — sometimes days — for an answer, and the answer arrives stitched together from three different sources that disagree with each other.

This system replaces all of that with one platform, three pieces:

1. The Squadron PC

Installed on each squadron's operations desk. Replaces the paper logbook and Excel files. Used by the Operations Officer to log every sortie.

2. The Pilot's Phone

A lightweight mobile companion. The pilot sees their own hours, currency, and upcoming flights. Pilots cannot edit official records — they only view.

3. The Secure Cloud

A hardened, encrypted database that quietly synchronises every PC and every phone in the background. Always up to date, always backed up.

What it changes for command

- One source of truth. What the squadron commander sees on their PC is what the wing commander sees on theirs — same numbers, same minute.
- Authority is enforced by the software, not by trust. A flight commander cannot accidentally — or deliberately — see another squadron's pilots unless you have explicitly authorised it.
- Paper forms still exist where regulation requires them (Risk Assessment, Duty Roster, Sortie Card) — but they are now generated by the system, in the exact official format, ready to print and sign.
- Nothing is ever lost. No more 'the laptop crashed and we lost March'. Every change is encrypted and replicated within seconds.
- Visibility between squadrons is governed by the licensing process at Headquarters. No subordinate, even with administrative access on their own PC, can grant themselves visibility over another squadron.

01 The Problem We Are Solving

Squadron operations today depend on three things that, individually, work — but together create real risk:

Paper Logbooks

Accurate when written, but locked inside one binder on one desk. A commander 200 km away cannot see them. If the binder is misplaced, damaged, or simply not updated for a week, command decisions are made on stale information.

Personal Spreadsheets

Every operations officer keeps 'their' Excel file. Format differs between squadrons. Numbers diverge. Sharing means emailing a copy — which is immediately out of date the moment the next sortie flies.

Phone & WhatsApp Updates

Currency status, who is on duty this week, who flew last night — passed verbally or on personal phones. Nothing is auditable. Nothing is preserved when a person rotates out.

No Enforced Boundaries

A spreadsheet emailed to one commander can be forwarded anywhere. There is no technical guarantee that information about Squadron 8 stays inside Squadron 8's chain of command.

> Operational consequence: When a wing commander needs an instant readiness number — for example, 'how many of our pilots are currency-current on type X tonight?' — the answer today requires phone calls to multiple squadrons and an hour of reconciliation. By the time it arrives, it may already be wrong.

02 The Solution — Three Pieces, One System

The system is intentionally split into three pieces because each piece has a different job, a different audience, and a different security profile.

Squadron Dashboard (Windows PC)

The official record. Operations Officer logs every sortie here. Replaces paper logbooks and Excel files.

Pilot Mobile App (iOS / Android)

Read-only personal view for each pilot. Pairs to the squadron PC by a one-time code.

Encrypted Cloud Sync

Hosted backbone (Supabase / PostgreSQL). All devices read and write through it. Backed up automatically.

How information flows

- The Operations Officer logs a sortie on the squadron PC.
- The PC encrypts the entry and sends it to the cloud over a secure channel.
- The cloud writes it to the squadron's protected slice of the database.
- Within seconds, every authorised PC and every paired pilot phone updates automatically — no email, no refresh button, no copying.

> Important: the Operations PC works fully offline as well. If the network drops, sorties keep being logged locally. The moment connection returns, the system catches up by itself. No data is lost.

03 Who Uses What — The Role Hierarchy

Every PC in the Force is licensed to a specific role at the moment its licence key is issued by the Headquarters licensing authority. The role is written into the key itself and cannot be changed locally by anyone — not even by an administrator sitting at that PC.

TIER 1 **Operations Officer**	Logs sorties, manages pilot list, generates risk assessments and duty rosters for their own squadron only. Cannot view any other squadron.
TIER 2 **Flight Commander**	All Operations capabilities, plus reporting and analytics across the squadrons explicitly authorised on this PC at the time the licence is issued.
TIER 3 **Squadron Commander**	Full visibility of their squadron's operations, currency, risk picture and rankings, plus any additional squadrons authorised on their PC.
TIER 4 **Wing / HQ Commander**	Multi-squadron command view. Authorised to monitor any combination of squadrons specified at issuance. Sees aggregated readiness and cross-squadron rankings.

> Key principle: Squadron-monitoring authority is never assigned to a person — it is assigned to a PC. When a commander rotates to a new appointment, the old PC's key is revoked and a new key is issued for the new PC. No backdoors. No drift.

04 Security — Explained Without Jargon

Every layer of this system was designed with the assumption that a determined adversary will eventually try to read it, copy it, or tamper with it. Below is what we do, and — more importantly — what each measure actually means in plain language.

1. Encryption in transit (TLS 1.3)

Every byte that leaves a squadron PC or a pilot's phone, on its way to the cloud, travels through an encrypted tunnel known as TLS (Transport Layer Security), version 1.3 — the most current internationally-recognised standard.

In plain language: imagine you are sending a letter from Amman to King Hussein Air Base. Without TLS, the letter is a postcard — anyone who handles it on the way can read it. With TLS, the letter is sealed inside an armoured diplomatic pouch that only the recipient holds the key to. Even if someone intercepts the pouch mid-route, all they see is unreadable noise.

2. Encryption at rest (AES-256)

When the data arrives at the cloud and is written to disk, it is encrypted again — this time using AES-256, the same standard used by NATO and most financial institutions for 'secret'-level material.

In plain language: imagine a bank vault. The pouch arrives at the bank; the contents are taken out and locked inside a vault. Even if someone walked away with the entire vault, they could not open it without the key — and the key is held separately, not in the vault itself.

3. Two-factor authentication (2FA) at first launch

Before any PC is allowed to talk to the system at all, an authorised operator must enter a one-time code generated by an authenticator app held by the Super Admin. Knowing the password alone is not enough.

In plain language: an ATM card by itself is useless. A PIN by itself is useless. You need both. 2FA gives every PC the same protection — the licence key alone is useless without the live one-time code.

4. Per-PC licence keys

Each PC receives a unique licence key tied to a specific role and a specific list of squadrons it is allowed to see. The key is generated by the Super Admin and activated once. After activation, the PC will refuse to talk to the cloud unless the key matches.

In plain language: think of it like a vehicle pass at a base gate. The gate guard checks the pass against the list. A pass for Squadron 8 will not get you into Squadron 12's hangar — and a forged pass will not even get you to the gate.

5. Per-squadron isolation (Row-Level Security)

Inside the database itself, every record is tagged with the squadron it belongs to. The database engine refuses to return any row unless the requesting PC's licence proves it is authorised to see that squadron. This is enforced at the deepest layer — even a software bug elsewhere in the system cannot leak data across squadrons.

In plain language: imagine a filing cabinet where every drawer has its own lock, and the cabinet itself refuses to even tell you a drawer exists unless your key opens it. You cannot accidentally open the wrong drawer because, from your perspective, the wrong drawer is invisible.

6. Tamper-resistant authority

Squadron-monitoring authority is set when a PC's licence key is issued and cannot be changed locally. A subordinate commander, even one with administrative access on their own PC, has no menu, no command, and no file path that will let them add a squadron to their own visibility list. The 'Manage Commanders' page on a commander's PC explicitly tells them: 'Which squadrons this commander can monitor is set centrally when the licence key for that PC is issued.'

7. Continuous, automatic backup

The cloud database is backed up automatically multiple times a day, in a geographically separate location. If a squadron PC is destroyed, lost, or stolen, no data is lost — a fresh PC with a freshly-issued key restores full operations within minutes.

8. Auto-update with verified releases

Every PC checks for new versions on launch and updates itself silently from the official release channel. This means a security fix reaches every squadron in the Force within hours, not months — and there is no chance of an outdated, vulnerable version lingering on a forgotten desk.

05 What Commanders Can Actually Do

Sortie Logging

Every flight recorded with date, mission type, aircraft, crew, duration, and remarks. The form mirrors the official sortie card so Operations staff are immediately at home with it.

Currency Tracking

The system continuously calculates each pilot's currency status against the 30-, 60-, 90- and 180-day windows defined in standing orders. Expiring currencies are flagged in amber; expired ones in red.

Risk Assessment Form

The official RJAF risk-matrix form is built into the system — identical layout, same threshold colours, prints onto two A4 pages ready to sign. No more hand-copying.

Squadron Duty Roster (bvDb63b†Hc’•

Full Arabic right-to-left weekly duty roster with Eastern-Arabic digits, automatic squadron name (e.g. bvDc61b, 'dF+bvEdb’Â –Æ÷B WFöf–ÆÂ ‡ ank + phone), pri commander-signature footer.

Pilot Rankings & Analytics

Top hours by month, by quarter, by aircraft type. Spotlight on under-utilised pilots. Helps commanders distribute flying load fairly and keep currency healthy across the squadron.

Monthly & Quarterly Reports

One-click PDF exports of squadron summaries, ready to brief at the next O-group. Numbers are pulled live from the same source the commander sees on screen.

Pilot Mobile View

Each pilot pairs their phone to the squadron PC with a one-time code. They see — read-only — their own hours, currency status, next-flight notice, and weekly duty.

Bilingual Interface

Full Arabic and English throughout. The interface mirrors correctly for RTL Arabic, including all forms, dropdowns, and printable documents.

06 What This Means for the Air Force

1. A real-time readiness picture

For the first time, a wing commander can answer the question 'how many qualified, currency-current pilots do I have on type X right now?' in seconds — and trust the answer.

2. Auditability and accountability

Every entry has a timestamp and a recording user. Disputed currencies, missing sorties, and 'I thought he was current' arguments disappear because the record is unambiguous and tamper-evident.

3. Continuity through personnel rotation

When an operations officer rotates out, no knowledge leaves with them. The new officer logs in on the same PC and finds the squadron's complete history, exactly where the previous officer left it.

4. Fair distribution of flying load

The rankings module makes over- and under-flown pilots immediately visible. Squadron commanders can correct imbalances before they become currency emergencies — or fairness issues.

5. Reduced administrative friction

Risk-assessment forms, duty rosters, and monthly reports — work that used to consume an evening at the end of every week — are now one-click outputs.

6. Force-wide standardisation

Every squadron, from training units to front-line, now uses the same form, the same fields, the same definitions. Cross-squadron analytics finally make sense because everyone is measuring the same things.

7. Resilience

Offline-tolerant on every PC. Encrypted at rest and in transit. Continuously backed up. Auto-updated. The system is engineered to keep operating through the conditions a squadron operations room actually faces.

8. Sovereignty of authority

Critically — and this is a deliberate design choice — squadron-monitoring authority cannot be altered locally on a commander's PC, by any user role, regardless of their administrative privileges on that PC. It flows only through the licence key issued by the Headquarters licensing authority. Authority lives in the chain of command, not in the software.

07 How It Is Operated Day-to-Day

Action	Who does it	Where
Issue a licence key for a new PC	HQ Licensing Authority	Licence Management !’ Generate Key
Adjust a commander PC's squadron scope	HQ Licensing Authority	Licence Management !’ 'Squadrons this PC monitors'
Revoke a PC's access	HQ Licensing Authority	Licence Management !’ Revoke
Log a sortie	Operations Officer	Squadron PC !’ Sortie Log
Generate weekly duty roster	Operations Officer	Squadron PC !’ Duty Week !’ Print
Generate risk assessment	Operations Officer	Squadron PC !’ Risk !’ Print
View own hours / currency	Pilot	Pilot's phone (paired to squadron)
Brief readiness to higher command	Squadron / Wing Commander	Their PC !’ Dashboard / Reports

What happens if a PC is lost or stolen?

- The lost PC's key is located in the central licence registry and revoked.
- On its next launch, the revoked PC is denied access to the cloud and cannot read any data, even with valid Windows credentials.
- A fresh PC is set up and a new key is issued; the new PC catches up automatically — no operational data was ever stored only on the lost machine.

08 Roadmap & Closing

Already shipped (current version)

- Per-squadron data isolation, enforced at the database layer
- Centralised squadron-monitoring control via licence-key issuance and live edit
- Arabic squadron duty roster with full RTL print
- Official RJAF risk-assessment form, two-page A4 print
- Mobile pilot pairing by one-time code
- TLS-1.3 in transit, AES-256 at rest, automatic backups
- Silent auto-update across the Force

Planned next

- Aircraft-by-aircraft maintenance status board
- Squadron schedule planner (next-day flying programme)
- Cross-squadron readiness heatmap for HQ
- Optional integration with national air-traffic and weather feeds

> Bottom line: the system is in active operational use, built around the way RJAF squadrons actually work, secured to international standards, and architected so that authority over who sees what stays exactly where it belongs — with command.

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